17 There are only 3 red sweets, 2 yellow sweets and 4 green sweets in a bag.

Nicola takes at random two sweets from the bag.

Work out the probability that the two sweets are **not** the same colour.

9 sweets initially; 8 after one is taken.

1 mark
Without replacement:
denominators9 and8.

P(RR) = (3/9) × (2/8) = 6/72

1 mark
Multiply along a matching-colour path.

P(YY) = (2/9) × (1/8) = 2/72

P(GG) = (4/9) × (3/8) = 12/72

P(same colour) = (6 +2 +12)/72 = 20/72

P(different colours) = 1 –20/72 = 52/72 = 13/18

1 mark
Add all THREE ways to take the same colour.

1 mark
Subtract from1 for colours that are not the same.

13/18

(Total for Question 17 is 4 marks)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NO